

RED SEAL EXAM PREP

306A Plumber

Free 50-Question Mock Exam

Practice for the Red Seal / Certificate of Qualification exam

Pass mark: 70% · Aligned to the published occupational standard topic weightings
<https://redsealquiz.ca> · Edition 2026-07

About This Material

Independent, unofficial study resource. This publication is not affiliated with, endorsed by, or approved by the Canadian Council of Directors of Apprenticeship (CCDA), Employment and Social Development Canada, the Red Seal Program, Skilled Trades Ontario, or any provincial or territorial apprenticeship authority. "Red Seal" is used solely to describe the examination for which this material provides practice.

Originality. All questions and explanations in this publication are original works created by the author. This material references only publicly available occupational standards (Red Seal Occupational Standard) for topic alignment and reproduces no content from any actual examination.

No guarantee of results. This study material is provided for educational and informational purposes only. Success on any certification examination depends on many factors, and no representation is made that using this material will result in passing any exam. Exam content, structure, and weightings are set by certification authorities and may change without notice.

Technical accuracy. Code values and technical specifications cited in explanations reflect commonly referenced Canadian standards at the time of writing. Always defer to the current edition of the applicable code or the manufacturer's documentation in real work.

Refunds. Because this is an instantly delivered digital file, all sales are final once the download link has been delivered. If you hit a technical problem — a corrupted or inaccessible file, or a duplicate purchase — contact us within 14 days and we will repair the file or refund you in full.

Updates. Corrections and additions are published to the same file. Re-download from your Ko-fi library to get the current edition at no extra cost.

© 2026 the author. All rights reserved. For personal study use by the purchaser only — redistribution, resale, or file sharing is prohibited.

Mock Exam — answer all questions, then check the key

Q1. What is the purpose of an overflow assembly in a bathtub?

- A) To connect the tub drain to the vent stack
- B) To prevent overflow if the drain is closed
- C) To allow draining the tub from both ends
- D) To increase drain capacity for a filled tub

Q2. ABS pipe joints are made using:

- A) Compression fittings with rubber O-rings
- B) ABS solvent cement with no separate primer
- C) Threaded connections cut into the pipe wall
- D) PVC purple primer followed by ABS solvent cement

Q3. A 306A plumber is checking a wet-vented bathroom group against the venting rules in the plumbing code. Which of these is NOT limited by the Code's wet venting provisions?

- A) The developed length of the wet vent pipe
- B) The number of wet-vented water closets
- C) The size of trap arms connected to the wet vent
- D) The length of a horizontal offset in the wet vent

Q4. What material is NOT approved for gas piping in residential construction?

- A) PVC plastic pipe, buried or exposed
- B) Corrugated stainless steel tubing (CSST)
- C) Copper tubing sized to the gas code tables
- D) Threaded black steel pipe, Schedule 40

Q5. A pipe wrench should NOT be used on which type of pipe?

- A) Cast iron pipe
- B) Black steel pipe
- C) Galvanized steel pipe
- D) Copper or plastic pipe

Q6. What is the purpose of dielectric unions when connecting copper pipe to galvanized steel pipe?

- A) To drop the supply pressure entering the older galvanized section of pipe
- B) To maintain electrical continuity across the joint for bonding of the water system
- C) To prevent galvanic corrosion between copper and galvanized steel
- D) To absorb the different thermal expansion rates of copper and steel piping

Q7. What is the function of a cleanout in a DWV system?

- A) To test drain slope during installation
- B) To allow water to drain from the pipe during freeze-up
- C) To give access for rodding out blockages
- D) To allow venting of sewer gas during maintenance

Q8. When threading black steel gas pipe, the correct pipe thread standard is:

- A) NPT (National Pipe Taper)
- B) BSP (British Standard Pipe)
- C) NPS (National Pipe Straight)
- D) Metric ISO threads

Q9. What is the purpose of the "flood level rim" reference in plumbing code?

- A) The fixture edge from which water would overflow
- B) The top of a water closet tank
- C) The maximum water level allowed in a river before flooding affects plumbing
- D) The maximum fill level of a water heater

Q10. What is the purpose of a P-trap in plumbing?

- A) To reduce water hammer in the drain line
- B) To filter debris out of the drain line
- C) To hold a water seal that blocks sewer gas
- D) To increase water pressure to the fixture

Q11. A 306A plumber wants to use a drainage arrangement that departs from the applicable acceptable solutions in Division B of the National Plumbing Code of Canada, and puts it forward as an alternative solution. Under that Code, who has to produce the supporting documentation, and what must it demonstrate?

- A) The plumber proposing it, showing it is adequate and safe in service
- B) The manufacturer of the components, in a written product approval
- C) The plumber proposing it, showing it performs as well as Division B
- D) The authority having jurisdiction, in the written review it prepares

Q12. What is a backflow preventer?

- A) A device preventing backflow into the potable supply
- B) A pressure reducer for high-pressure zones
- C) A check valve that prevents water hammer
- D) A valve that prevents hot and cold water from mixing

Q13. Where must a main water supply shutoff valve be located?

- A) At the street property line
- B) At the outdoor water meter location only
- C) In the basement only
- D) Inside the building at the service entry

Q14. What is the difference between high-pressure and low-pressure gas distribution?

- A) High pressure serves commercial buildings; low pressure serves homes
- B) High pressure uses CSST only; low pressure uses black steel only
- C) A single pressure is used throughout the whole distribution system
- D) High pressure serves street mains; low pressure serves appliances

Q15. What is a potable water system cross-connection?

- A) A potable water line that crosses from one building to another over a property line
- B) Any tee fitting where a branch line is taken off the potable water distribution main
- C) A connection between potable water and a contamination source
- D) A point where the hot and cold supply lines are joined ahead of a mixing valve

Q16. What is the purpose of a wax ring when installing a water closet?

- A) To secure the toilet to the floor structurally
- B) To absorb vibration from the toilet
- C) To align the toilet with the drain
- D) To seal the toilet horn to the closet flange

Q17. What is the purpose of the authority having jurisdiction (AHJ) in plumbing inspections?

- A) To train plumbing apprentices
- B) To certify plumbing products and materials
- C) To inspect work and enforce the plumbing code
- D) To sell plumbing permits to homeowners

Q18. What is the purpose of a temperature and pressure (T&P) relief valve on a water heater?

- A) To limit the hot water delivered to the fixtures to a safe tap temperature
- B) To absorb thermal expansion in a closed system so the tank is not overstressed
- C) To drop incoming cold water pressure down to the heater's rated working pressure
- D) To release water automatically on excess pressure or temperature

Q19. A vent pipe is carried up through the roof of a single-storey wing of a building and terminates in outside air. In the wall of the taller section beside it, an openable window has its head at the same height as that vent terminal. Under the National Plumbing Code of Canada, what minimum horizontal distance is required between the terminal and the window?

- A) 3.5 m
- B) 2 m
- C) 1 m
- D) 1.8 m

Q20. Under the National Plumbing Code of Canada, what is the maximum water usage per flush cycle for a new single-flush water closet in a residential occupancy?

- A) 4.8 L (1.28 gpf)
- B) 3.8 L (1.0 gpf)
- C) 13.2 L (3.5 gpf)
- D) 6.0 L (1.6 gpf)

Q21. A hot water heater is producing discoloured (rusty) water but the cold water is clear. What should the plumber suspect FIRST and recommend?

- A) Water softener malfunction — check salt levels
- B) Hot water piping corrosion — replace all copper piping
- C) The water main has a main break upstream — flush the system
- D) The anode rod has depleted — replace the sacrificial anode

Q22. A 306A plumber inspects a suite that has stood unoccupied for three weeks and finds that a fixture trap has lost its water seal. The fixture has not been used, no other fixture on the branch has been used, the branch is properly vented, and the trap and its connections are dry and sound. What is the MOST likely explanation?

- A) Back pressure in the stack has blown the trap seal out
- B) The discharge of an adjacent fixture siphoned the seal out
- C) Capillary action has drawn the seal out over a wick of lint
- D) The seal has evaporated over the long period of disuse

Q23. What are the two main fuel gases supplied to homes in Canada?

- A) Hydrogen and methane
- B) Methane and propane
- C) Acetylene and propane
- D) Butane and natural gas

Q24. What is the purpose of a drip leg (sediment trap) in a gas line?

- A) To reduce gas pressure to the appliance
- B) To trap condensate and debris before the appliance
- C) To serve as a shutoff valve location
- D) To equalize pressure between high and low pressure gas

Q25. What is a dual-flush toilet and what is its advantage?

- A) A toilet that can be flushed from either side of the tank
- B) A toilet with two tanks for a higher flush volume
- C) A toilet offering a reduced flush and a full flush
- D) A toilet with both siphon-jet and washdown action

Q26. When using a pipe threading machine, the correct lubricant to use is:

- A) Motor oil
- B) Water
- C) Threading (cutting) oil
- D) No lubricant needed for power threading

Q27. What is the minimum slope (grade) for a 3-inch (75 mm) horizontal drain pipe?

- A) No slope required for drain pipes
- B) 2% (1/4 in per foot)
- C) 4% (1/2 in per foot)
- D) 1% (1/8 in per foot)

Q28. What is the minimum insulation required for hot water supply pipes in an unconditioned space?

- A) Insulation required only for pipes exceeding 75 mm diameter
- B) No insulation required
- C) The thermal resistance required for exterior walls
- D) Insulation is only required for pipes in exterior walls

Q29. What is a lockout/tagout (LOTO) procedure used for in plumbing work?

- A) To lock plumbing fixtures during inspection
- B) To mark pipe sizes during installation
- C) To tag pipes for pressure testing
- D) To isolate energy sources before maintenance

Q30. Why are S-traps not accepted as fixture traps in modern plumbing?

- A) S-traps self-siphon and lose their seal
- B) They require too much head room below the fixture
- C) They are too expensive to install and maintain
- D) They are not compatible with modern ABS pipe

Q31. A stack in a 6-storey apartment building drains fixtures on every storey and does not serve as a wet vent, so the plumbing code requires a vent stack for it. Where must the lower end of that vent stack be connected?

- A) To a vertical section of the stack, immediately above the highest drainage connection to it
- B) To a vertical section of the stack, at least 1 m above the floor level of the lowest storey
- C) To the horizontal building drain, immediately downstream of the base of the stack served
- D) To a vertical section of the stack, at or immediately below the lowest drainage connection

Q32. A 306A plumber is replacing a section of hot water piping in a building put up in the 1960s. The pipe is lagged with a hard, plaster-like insulation jacket that will have to be cut through, and no report on the building's materials has been provided. What has to happen before that insulation is disturbed?

- A) Wet it down and cut it out with hand tools
- B) Judge it from its look and the building's age
- C) Disturb only the short length the repair needs
- D) Have the material tested before disturbing it

Q33. A 306A plumber is fitting a bathroom exhaust fan in a dwelling unit. The principal ventilation fan's exhaust air intake is elsewhere in the house, so this bathroom needs a supplemental exhaust fan. The duct run is 7 m with two 90° elbows. Under NBC 2020 / OBC 2024 Article 9.32.3.7, what minimum rated capacity applies, and what still has to be checked?

- A) 10 L/s — a bathroom is exempt from the ventilation calculation
- B) 25 L/s — then confirm delivered airflow at system static pressure
- C) 50 L/s — the same figure the Code sets for a kitchen fan
- D) No minimum — bathroom exhaust follows the fan maker's advice only

Q34. A new residence has copper water pipe that shows blue-green staining on fixtures and pinhole leaks after 2 years. What is the MOST likely cause and recommended solution?

- A) Type M copper was used — upgrade to Type L
- B) Acidic (low pH) water is corroding the pipe — adjust pH or replace pipe
- C) Copper pipe was installed in direct contact with concrete
- D) The joints were soldered with 95/5 solder instead of lead-free solder

Q35. A domestic clothes washer is being roughed in and will discharge into its own trap rather than into a laundry tray. What does the plumbing code require of the vertical standpipe fitted to that trap inlet?

- A) At least 600 mm from the trap weir, ending above the washer's flood level rim
- B) At least 450 mm from the trap weir and not more than 1 070 mm above the weir
- C) At least 300 mm from the trap weir, ending below the washer's flood level rim
- D) Any height, provided the trap arm stays within its permitted developed length

Q36. What is a floor drain trap primer?

- A) A chemical treatment used to kill bacteria in floor drain traps
- B) A device that automatically adds water to maintain the trap seal
- C) A primer applied to the drain pipe before installing the trap
- D) A backflow preventer fitted on the outlet of the floor drain

Q37. What is the correct procedure before opening a gas line for work?

- A) The line self-purges when opened
- B) Shut off the gas, purge the line, and test
- C) Open windows for ventilation only
- D) Work quickly to minimize gas release

Q38. What is the maximum length of a trap arm (distance from trap weir to vent)?

- A) Varies by pipe diameter — set by NPC venting tables
- B) Unlimited as long as the required 1 in 50 slope is maintained throughout
- C) Fixed at 600 mm (24 in) from the trap weir for every pipe size
- D) One foot of developed length for each inch of trap arm diameter

Q39. What is the purpose of an air admittance valve (AAV)?

- A) To admit outside air into a septic system
- B) To supply air to the combustion chamber of a water heater
- C) To bleed air from pressurized water supply lines
- D) To admit air into the drain to stop siphonage

Q40. When is a sanitary drainage system required to have an interceptor other than a grease trap?

- A) When the drainage velocity in the branch exceeds 1 metre per second
- B) When the waste could damage or impair the drainage system
- C) Only where the building stands in an industrial zoning district
- D) In every building that has more than five plumbing fixtures

Q41. What is the standard leak test for gas piping after installation per CSA B149.1?

- A) Leak detection spray at fittings only
- B) Visual inspection of joints only
- C) Pressure test with air or nitrogen
- D) Fill with water and check for leaks

Q42. Why is a short length of metal pipe or an approved connector fitted between a water heater and PEX tubing?

- A) PEX has a lower working pressure rating than copper
- B) Heat at the heater outlet can exceed the tubing rating
- C) PEX must be bonded to the electrical grounding system
- D) PEX cannot be joined to threaded fittings of any kind

Q43. What is a wet vent?

- A) A vent pipe that admits outside air to the building drain
- B) A vent pipe fitted with a trap to seal off sewer gas
- C) A single vent serving several fixtures on one branch
- D) A soil-or-waste pipe that also vents other fixtures

Q44. What is the difference between a sanitary sewer and a storm sewer?

- A) Sanitary sewer is for commercial buildings; storm sewer is for residential
- B) They are the same in modern buildings
- C) Sanitary sewer carries sewage; storm sewer carries surface water
- D) Sanitary sewer is deeper; storm sewer is shallower

Q45. How is the minimum size of a vent pipe serving a fixture determined?

- A) From the size of the trap served, using the code table
- B) As one third of the drainage pipe that the vent serves
- C) As one half of the drainage pipe that the vent serves
- D) The same size as the drainage pipe that the vent serves

Q46. What is the minimum hot water temperature at the water heater to prevent Legionella growth?

- A) 45°C (113°F)
- B) 70°C (158°F)
- C) 80°C (176°F)
- D) 60°C (140°F)

Q47. What is a condensate drain line used for in HVAC/plumbing?

- A) To collect rainwater from condensing boilers
- B) To drain water softener backwash
- C) To drain pool water from above-ground swimming pools
- D) To drain condensate from AC and high-efficiency furnaces

Q48. A new potable water distribution system has passed its pressure test. What must be done before the building is occupied?

- A) Clean, flush and sanitize the system before occupancy
- B) Rely on the chlorine already in the municipal supply
- C) Run every fixture until the water is clear and odourless
- D) Add a scale inhibitor chemical to the potable lines

Q49. What is required when a plumber discovers unapproved (non-permitted) plumbing work during a renovation?

- A) Proceed with the renovation and do not report it
- B) The plumber must refuse all work on the property
- C) Inform the owner and recommend a permit and inspection
- D) Nothing — historical non-permitted work is grandfathered

Q50. What is the primary hazard when cutting galvanized pipe with a torch?

- A) Risk of electric shock
- B) Zinc oxide fumes causing metal fume fever
- C) Pipe collapse from heat
- D) Carbon monoxide from acetylene combustion

Answer Key & Explanations

Q1. Answer: B

The overflow assembly provides an emergency drain opening at the upper end of the tub. If the drain is blocked or closed and the tub fills, water enters the overflow, preventing spillage onto the floor.

Q2. Answer: B

ABS drainage pipe is joined with a one-step ABS solvent cement. The cement itself softens both surfaces and they fuse as it cures, so no separate primer step is used. PVC is the material that needs the two-step treatment: primer first to soften the surface, then PVC cement. Carrying the purple-primer habit over from PVC does not improve an ABS joint and is not the ABS procedure. ABS is a solvent-weld material, so a DWV run is not made up with compression fittings, and plastic DWV pipe wall is not designed to carry cut threads.

Q3. Answer: A

The wet venting article limits nearly everything about a wet vent except how far it runs. A soil-or-waste pipe may serve as a wet vent provided the hydraulic load is within the venting load table; not more than 2 water closets are wet-vented, and where there are two they are connected at the same level and installed downstream of all the other fixtures; trap arms and fixture drains connected to the wet vent do not exceed 2 inches; there is not more than one nominally horizontal offset, and that offset does not exceed 1.2 m for pipe 2 inches or smaller (2.5 m for larger pipe); and the wet-vented portion is not reduced in size. The article then closes by stating in as many words that the length of the wet vent is not limited. That matters on the job: a wet vent cannot be failed on run length alone, so any argument about one has to be made on hydraulic load, fixture count, pipe size or offset. The usual source of an invented wet vent length limit is the trap arm, whose maximum length is set by its own table — that limit belongs to the trap arm and does not transfer to the wet vent.

Q4. Answer: A

PVC is not a fuel gas piping material. The gas installation code's materials section recognizes steel pipe, copper tube, steel tube, polyethylene, corrugated stainless steel tubing, corrugated metal connectors and hose connectors, and it requires that all plastic gas pipe, tube and fittings be polyethylene complying with CSA B137.4. A material the code does not name may be used only where it conforms to a nationally recognized standard or to a test report of a nationally recognized certification organization, and PVC has no such standing for fuel gas. That is exactly where the two plastics get confused: there is a plastic gas pipe in the trench, but it is polyethylene, and polyethylene is approved only for outdoor underground work - distribution mains, house service lines, and on-premise lines past the meter. Steel pipe is the common above-ground material, threaded, conforming to ASTM A53/A53M or A106 in Schedule 40, Schedule 80 or standard weight, and hot dipped zinc coated pipe and fittings may be used. Copper is on the list too, and the old shop rule that copper is banned for propane is not what the gas code says: the code covers natural gas and propane together, and Type G copper tube is made to ASTM B837, 'Standard Specification for Seamless Copper Tube for Natural Gas and Liquefied Petroleum (LP) Gas Fuel Distribution Systems'. Copper for gas is conditional - Types G, K and L above ground provided they do not exceed 1-3/8 in OD, and soft annealed Type K or externally coated tube underground - so a copper question is settled against the gas code in force and the authority having jurisdiction, not by a blanket rule either way.

Q5. Answer: D

Pipe wrenches are designed for steel pipe. Using them on copper or plastic pipe crushes the softer material, causing leaks and damage.

Q6. Answer: C

Copper and galvanized steel (zinc-coated iron) form a galvanic cell when in contact with water, accelerating corrosion of the steel (zinc sacrificial anode effect). Dielectric unions isolate the two metals.

Q7. Answer: C

Cleanouts give drain cleaning tools - rods, snakes, hydro-jet - a way into the piping to clear blockages. The Code sets where they go: at the bottom of a soil-or-waste stack, or within 3 m upstream of it, or on a Y fitting at the building drain; as close as practical to where the building drain leaves the building; on the upstream side and directly over every running trap; downstream of an interceptor; and at the maximum spacing given in the cleanout table.

Q8. Answer: A

Gas piping uses NPT (National Pipe Taper) threads, which create a pressure-tight seal as the tapered male and female threads engage. Must use yellow PTFE tape or pipe thread compound rated for gas.

Q9. Answer: A

The flood level rim (FLR) is the top edge of a fixture from which water would overflow (top of a sink bowl, edge of a bathtub). Code uses FLR to set minimum fixture placement height and supply/trap requirements.

Q10. Answer: C

The P-trap retains a water seal, a trap seal depth of not less than 38 mm, that blocks sewer gas including hydrogen sulphide and methane from entering the living space through the drain.

Q11. Answer: C

The burden sits with whoever asks for the alternative solution: the plumber proposing it must document that it performs at least as well as the Division B acceptable solutions it departs from. The National Plumbing Code of Canada 2020 gives two routes to compliance and no others - complying with the applicable acceptable solutions in Division B, or using alternative solutions that "will achieve at least the minimum level of performance required by Division B in the areas defined by the objectives and functional statements attributed to the applicable acceptable solutions". Division C then puts the paperwork where the proposal came from: documentation "shall be provided by the person requesting the use of an alternative solution" to demonstrate that it complies, and it must include a Code analysis of the analytical methods and rationales used to determine that the proposal reaches that level of performance, plus the qualifications, experience and background of whoever takes responsibility for the design. Wrong answers: arguing that the arrangement is adequate and safe in service is the commonest error, and the Code's own explanatory note shuts it down - the test is not "well enough" but "as well as" a design that would satisfy the applicable acceptable solutions. The authority having jurisdiction is where that documentation is submitted and where it is retained, not where it is written; the Code never asks the regulator to make the proponent's case. A manufacturer's product approval speaks for a component's own performance and says nothing about whether an arrangement that departs from Division B reaches the performance Division B sets.

Q12. Answer: A

Backflow preventers protect the potable water supply from contaminated water flowing backward into the system under back-siphonage or back-pressure conditions. Required at cross-connections.

Q13. Answer: D

NPC requires a main shutoff valve inside the building at the point where the water service enters, accessible at all times for emergency shutoff of water to the entire building without specialized tools.

Q14. Answer: D

Gas leaves the utility's supply mains at a pressure far above anything an appliance can accept, and it is stepped down in stages on the way to the burner. A service regulator at the meter, installed by the utility on a natural gas system, reduces the supply to the building's line pressure. Where that line pressure is still above what the appliance is rated for, a line pressure regulator brings it down to the 0.5 psig (14 in WC) maximum inlet pressure most residential and light commercial appliances accept. The appliance's own regulator then sets burner manifold pressure, roughly 3 to 4 in WC for natural gas and 10 to 11 in WC for propane. Pressure class does not decide the piping material, and commercial and residential buildings are fed from the same high-pressure mains.

Q15. Answer: C

A cross-connection is any actual or potential physical connection between the potable water supply and a non-potable source of contamination (irrigation systems, chemical tanks, sewer, etc.). Cross-connections must have appropriate backflow prevention.

Q16. Answer: D

The wax ring creates a flexible, watertight seal between the toilet's horn (outlet) and the closet flange, preventing sewer gas and sewage leaks at the floor connection.

Q17. Answer: C

The AHJ (typically a municipal building department or provincial authority) enforces the adopted plumbing code by reviewing permit applications and plans, issuing permits, inspecting installations, and issuing compliance certificates.

Q18. Answer: D

The T&P relief valve carries two sensing elements in one body. The temperature element sits in the top of the tank and opens to keep the water from exceeding 99 C; the pressure element opens when the pressure in the tank reaches the tank's rated working pressure. Either condition on its own can rupture a heater, so the valve discharges water rather than let the vessel fail. Its discharge pipe must be at least the size of the valve outlet, rigid, sloped downward, unthreaded at the outlet, and terminated by an indirect connection above a floor drain or other safe location, with no shut-off valve anywhere on the line.

Q19. Answer: A

A vent terminal has to stay clear of anything that could draw sewer gas back into a building, and the Code puts all of those clearances in one sentence. Except for a fresh air inlet, a vent pipe terminated in outside air is to be located not less than 1 m above and not less than 3.5 m in any other direction from every air inlet, openable window or door; not less than 2 m above and not less than 3.5 m in any other direction from a roof that supports an occupancy; not less than 2 m above ground; and not less than 1.8 m from every property line. The 1 m figure buys clearance straight up only. Here the window head is level with the terminal, so the terminal is not above the opening and the clearance required in any other direction governs: 3.5 m. The three losing figures are all real, but each belongs somewhere else in the same sentence - 1 m is the vertical clearance where a vent is carried up past the opening, 2 m is the clearance above ground and above an occupied roof such as a sun deck, and 1.8 m is the setback from a property line. Two further requirements travel with this terminal: it must be not less than 150 mm above the roof or above storm water that could pond on the roof, and it must be flashed where it passes through the roof.

Q20. Answer: A

Table 2.6.1.6. of the National Plumbing Code caps a residential water closet at 4.8 L per flush cycle. The 6.0 L figure is the limit for industrial, commercial and institutional water closets, and Sentence 2.6.1.6.(4) allows 6.0 L for a single-flush closet in a residential retrofit only where 4.8 L can be shown to be impracticable given the existing building or municipal infrastructure. A dual-flush unit rated 6.0 L and 4.1 L or less is deemed to comply. Urinals are limited to 1.9 L. Toilets installed before the 1990s used 13 L or more per flush, and 3.8 L (1.0 gpf) is a manufacturer's ultra-high-efficiency rating, not a code limit.

Q21. Answer: D

Rusty hot water with clear cold water indicates tank corrosion — the sacrificial anode rod in the hot water tank has failed. Magnesium or aluminum anode rods protect the glass-lined steel tank through galvanic action. Once depleted, the tank corrodes internally. Replacing the anode rod early extends tank life. If the tank interior is already significantly corroded, replacement of the whole tank may be more cost-effective.

Q22. Answer: D

A trap seal lost with no use, no blockage and no leak has evaporated. Traps are required to have a trap seal depth of not less than 38 mm, and 50 mm on fixtures draining to an acid waste system. A seal that deep does not disappear overnight, but in a warm dry building an unused trap can lose it over a few weeks - which is why the code requires that provision be made for maintaining the trap seal of a floor drain, by a trap seal primer, by using the drain as the receptacle for an indirectly connected drinking fountain, or by other equally effective means. A deep-seal trap or scheduled manual refilling serves the same end on other fixtures. The three wrong answers all need something the scenario rules out: siphonage and back pressure both need a discharge somewhere in the system to drive them, and capillary action needs a wick of lint or hair bridging the trap.

Q23. Answer: B

Natural gas distributed through municipal pipelines is essentially methane (CH₄). Propane (LP gas/LPG) is delivered as a liquid to a cylinder or tank and serves homes beyond the reach of the gas mains. The two are separate fuels, not two grades of one — Canada's installation code is titled for both natural gas and propane, and an appliance is orificed and regulated for one or the other.

Q24. Answer: B

A drip leg is a vertical pipe extension below the horizontal run, installed just before the gas appliance. It collects condensate, debris, and scale from the gas supply before they reach the appliance regulator or valve.

Q25. Answer: C

A dual-flush toilet has one flush valve with two actuators: a reduced flush for liquid waste and a full flush for solids, which lowers average water use over the life of the fixture. Table 2.6.1.6. of the National Plumbing Code caps a residential water closet at 4.8 L per flush cycle and an industrial, commercial or institutional water closet at 6.0 L, and a water closet with a dual flush cycle of 6.0 L and 4.1 L or less is deemed to comply. Older single-flush toilets used 13 L or more per flush. Wrong answers: a dual-flush toilet has one tank, not two, and its purpose is to reduce volume rather than raise it; the two buttons operate a single valve, so the fixture is not flushed from one side or the other; and siphon-jet versus washdown describes bowl and trapway design, which is a separate matter from how many flush volumes the fixture offers.

Q26. Answer: C

Threading oil (cutting oil) lubricates and cools the die head, flushes the chips out of it, and lets the dies form a clean, full thread - which is what makes a tight, leak-free joint and what keeps the dies sharp. Wrong answers: motor oil is a hydrodynamic lubricant with no extreme-pressure sulfur or chlorine additives, so it will not stop the die teeth from welding and galling to the pipe, and it does not carry chips out of the die head - the threads come out torn. Water flash-rusts both the pipe and the dies and provides no boundary lubrication at all. Running a power threader dry is not an option either: the dies overheat within a few threads, the thread form is destroyed and the die set is ruined. Dark or black threading oil is the general-purpose product; clear or 'white' threading oil is used on stainless steel and on galvanized or potable-water work.

Q27. Answer: B

A 75 mm (3 in) horizontal sanitary drain is graded at 2% — 1 in 50, or 20 mm per metre, which is 1/4 in per foot. With too little grade the flow is too slow to carry solids along with it; with too much grade the liquid races ahead and leaves the solids behind. Note which way the rule runs: larger-diameter drains are permitted a flatter grade than smaller ones, because a bigger pipe carrying the same discharge still develops the depth and velocity needed to scour the invert. Do not assume the 2% figure applies to every size — read the grade off the table in the code in force for the pipe size you are running.

Q28. Answer: C

The National Building Code's energy efficiency provisions set this by comparison rather than by a fixed R-value: piping outside the building envelope or in an unconditioned space must be insulated to a thermal resistance not less than the effective thermal resistance required for exterior above-ground walls, so the number moves with the climate zone. Two related rules sit beside it — the first 2 m of inlet and outlet piping at a storage tank takes insulation at least 12 mm thick, and so does all the piping of a continuously recirculating system. There is no R-1 minimum anywhere in Canadian code; R-1 to R-3 pipe minimums belong to US energy codes.

Q29. Answer: D

LOTO ensures energy sources — water, gas, and electrical — are isolated and de-energized before performing maintenance. In plumbing, this includes closing and locking valves and tagging them to prevent accidental re-energization.

Q30. Answer: A

An S-trap has both an upward and a downward dip, so its outlet turns straight down. The discharging water carries on down that second curve and pulls the water seal with it — the trap siphons itself dry and stops blocking sewer gas. The code reaches the same result by limiting the total fall of a trap arm to not more than the arm's inside diameter, which an S-trap cannot meet.

Q31. Answer: D

A vent stack is required where a stack drains fixtures from more than 4 storeys containing plumbing fixtures, unless that stack serves as a wet vent, and the Code fixes where the pipe starts: it is connected to a vertical section of the stack at or immediately below the lowest sanitary drainage pipe connected to that stack. The definition of the pipe says the same thing from the other end - a vent stack is connected at its lower end to the stack at or below the lowest sanitary drainage pipe connection, and at its upper end either to a vent header or to outside air. Taking the lower end off the horizontal building drain past the foot of the stack does not satisfy that Article, which calls for a vertical section of the stack itself; the model codes used south of the border do allow a building-drain connection, which is why the idea travels. Connecting immediately above the highest drainage connection describes the other pipe entirely: the stack carried on past its highest drainage connection is the stack vent, and the Code requires the upper end of every stack to terminate in one. That "at or above the highest sanitary drainage pipe connection" wording does exist a few Articles later, but it governs the second offset relief vent on an offset stack, not the base of a vent stack. The 1 m dimension belongs to yoke vents: on a stack receiving discharge from fixtures on more than 11 storeys, the yoke vent connects to the vent stack at least 1 m above the floor level of the lowest storey of the section it relieves. Carrying that figure down to the vent stack's own connection puts the fitting a storey too high and leaves the base of the stack unrelieved.

Q32. Answer: D

Insulation of that age and description is suspect until a laboratory says otherwise, and identification comes before disturbance, not after it. CCOHS puts it plainly: you cannot tell whether a product contains asbestos just by looking at it, and the only way to tell is to have it tested by a laboratory. The sample is arranged through the building owner or the employer before the work starts, and if asbestos is confirmed the job becomes an asbestos operation with its own procedures, containment, training and disposal under the occupational health and safety legislation of the province. Judging the material by its appearance and the age of the building is the failure that puts fibres in the air. Age is worth noticing - it is what raises the suspicion in the first place - but it settles nothing either way, and a plumber who guesses wrong has already released the fibres by the time anyone discovers the error. Wetting the material down and working it with hand tools is a control that belongs inside an assessed procedure for material that has already been identified; reaching for it first substitutes a control for the assessment that decides which controls apply. Keeping the disturbed length short does not help either, because the hazard is airborne fibre: the release happens at the first cut, and a small opening in friable insulation contaminates the room and everyone in it. If suspect material turns up once work is already under way, the answer is the same - stop, leave it alone, and have the area secured until it has been identified.

Q33. Answer: B

NBC 2020 / OBC 2024 Article 9.32.3.7.(4): where the principal ventilation fan's exhaust air intake is not in the bathroom or water-closet room, a supplemental exhaust fan of rated capacity not less than 25 L/s shall be installed there. The Code states the figure in litres per second only, and the same Article sets 50 L/s for a kitchen, and mixing the two up is the classic error. Watch the imperial conversion: 25 L/s is about 53 CFM converted properly, but the requirement is widely published as "50 cfm" using the rough 1 L/s = 2 CFM convention, so a fan rated exactly 50 CFM sits right on the line. Size above it rather than at it. Rated capacity is also not delivered capacity: published ratings are taken at a set external static pressure, and 7 m of duct with two 90° elbows adds resistance, so check the fan curve against the real duct run. Two figures that are NOT the Canadian requirement and are easy to import by mistake: ASHRAE 62.2's 50 CFM intermittent / 20 CFM continuous is a US standard with no force here, and the 23 L/s intermittent / 9 L/s continuous pair was a British Columbia-only amendment that BC dropped when it harmonized to the national wording.

Q34. Answer: B

Blue-green staining on fixtures indicates copper dissolution — the water is attacking the pipe from inside. Acidic water (pH below 6.5) aggressively corrodes copper, causing blue-green deposits on fixtures and eventually pinhole leaks. Solutions: pH adjustment (neutralizing filter), water softener bypass, or replace with plastic pipe (PEX, CPVC) that resists acidic water. Report to local water authority.

Q35. Answer: A

Where clothes washers do not drain to a laundry tray, the trap inlet shall be fitted with a vertical standpipe that is not less than 600 mm long measured from the trap weir, and the top of the standpipe shall terminate above the flood level rim of the clothes washer it serves. Both halves of that Sentence carry weight. The 600 mm of vertical pipe gives the washer's discharge pump somewhere to put a surge of water arriving faster than the trap can pass it, and terminating above the flood level rim is what keeps that surge from spilling out of the standpipe onto the floor. The same table that sets fixture outlet pipe sizes lists a domestic clothes washer at 2 fixture units with a 2 in. trap. Wrong answers: the Canadian Article states a minimum length only, so a stated minimum-and-maximum band above the weir is not how the requirement is written; a standpipe stopped below the washer's flood level rim overflows under exactly the condition it exists to absorb; and the developed length limit belongs to the trap arm, the horizontal run from the trap weir to the vent, and says nothing about how high the pipe rises above the trap. The height is not unlimited either, but the ceiling comes from a separate rule - the developed length of a fixture outlet pipe shall not exceed 1 200 mm.

Q36. Answer: B

Floor drain traps can dry out (evaporate) if seldom used. A trap primer automatically supplies a small amount of water to maintain the seal, preventing sewer gas entry.

Q37. Answer: B

Before opening any gas line: shut off at the meter or upstream valve, purge the line of remaining gas, and use a calibrated gas detector to verify the atmosphere is below 10% LEL before any ignition sources.

Q38. Answer: A

The maximum trap arm length (distance from trap weir to the vent connection) is specified in NPC tables based on pipe diameter. Exceeding this causes trap siphonage.

Q39. Answer: D

An air admittance valve opens to admit air when negative pressure develops in the drain, preventing trap siphonage without running a vent pipe from that fixture up through the roof. It closes under positive pressure so sewer gas cannot escape through it.

Q40. Answer: B

An interceptor is triggered by what the waste contains, not by the size, velocity or zoning of the installation.

The general rule is that where a fixture or equipment discharges sewage or waste that may damage or impair the sanitary drainage system, or the functioning of a public or private sewage disposal system, provision must be made to treat that waste before it is discharged. The specific requirements outside grease are an oil interceptor where the discharge from a fixture may contain oil or gasoline - service garages, vehicle repair bays, fuelling stations - and a purpose-designed interceptor where a fixture discharges sand, grit or similar materials, as at a car wash or machine shop. Fixture count and drainage velocity trigger nothing.

Q41. Answer: C

Gas piping is proved by pressurizing it with air or an inert gas such as nitrogen or carbon dioxide. The fuel gas itself is never used as the test medium, and oxygen is never used because it can ignite on contact with oil. Test pressure and minimum hold time are read from the pressure test table in CSA B149.1, which sets them by the system's working pressure, size and length rather than by any fixed multiplier: a 2 psi system 50 m long is held at 15 psi for at least 15 minutes, while the same system 70 m long must hold for 60 minutes. This test is done before any appliance is connected, with components rated below the test pressure, such as line pressure regulators, isolated or removed. Soap solution is used afterward to locate a leak the pressure drop has already revealed.

Q42. Answer: B

PEX is certified to a maximum continuous service temperature, and the metal outlet fitting of a storage water heater can sit hotter than the water leaving the tank — on a gas-fired heater it is also close to the flue passage. A short metal pipe, or a connector listed for use on the appliance, takes that heat, and the run transitions to PEX beyond it. How far out the transition has to sit is set by the tubing certification and the appliance installation instructions, so there is no single distance that applies everywhere. PEX is joined to threaded fittings routinely using transition adapters, so that is not the reason. It is non-conductive and is not used as part of the electrical bonding path. And its pressure rating is not what limits it at the tank — temperature is.

Q43. Answer: D

A wet vent is a soil-or-waste pipe that the code permits to serve as a vent as well. It carries the discharge of the fixtures upstream of it and at the same time vents fixtures connected to it, so one pipe does two jobs and no separate dry vent is run to those fixtures. The code allows the arrangement only where the conditions set out in its wet venting article are all met, and the pipe is sized from the hydraulic load draining into it using the venting load table rather than by rule of thumb. The wrong answers are each a different thing with its own name. A single vent serving several fixtures on one nominally horizontal branch is a circuit vent, which the code sets out in its own article separate from wet venting. A vent fitted with a trap is not a code arrangement at all: the crown-vented trap, the closest thing to it, is prohibited. And a pipe that admits outside air to the drainage system near the building trap is a fresh air inlet, another of the vent types the code lists alongside the wet vent.

Q44. Answer: C

Sanitary sewers carry domestic and commercial sewage and wastewater to treatment plants. Storm sewers collect rain and surface runoff, often discharging directly to waterways. In modern construction, they must be kept separate (separated sewer).

Q45. Answer: A

Vent size comes from the trap, not from a fraction of the drain. The code table of minimum permitted vent sizes is entered with the size of the trap served: a 1-1/4 in. trap takes a 1-1/4 in. vent, a 1-1/2 in. trap also takes 1-1/4 in., a 2 in., 3 in. or 4 in. trap takes a 1-1/2 in. vent, and a 5 in. or 6 in. trap takes a 2 in. vent. So a large trap does not produce a proportionally large vent, and no fixed fraction of the drain is used anywhere in the venting rules for fixture vents. The one fraction that does appear applies to stacks, not to fixture drains: a vent stack or stack vent is not less than one-half the size of the soil-or-waste stack at its base. A branch vent, stack vent, vent stack or vent header is also never smaller than the vent pipe it connects into.

Q46. Answer: D

Legionella grows vigorously in warm water, roughly 25°C to 45°C, and is killed by prolonged exposure above 60°C. That is why a storage water heater is kept at 60°C rather than at a comfortable delivery temperature, and why the code fixes the thermostat setting for an electric storage-type service water heater at 60°C. A thermostatic mixing valve then blends the stored water down so that what reaches the fixture cannot scald. Note the direction of each figure: 60°C is a floor in the tank, and the delivery figure is a ceiling at the fixture.

Q47. Answer: D

Air conditioners and high-efficiency condensing furnaces produce significant condensate water. The condensate drain line (typically small-diameter PVC or vinyl tubing) routes this water to a floor drain, sump, or utility sink.

Q48. Answer: A

Commissioning a potable system is more than a pressure test. The lines are first flushed to carry out construction debris, cutting oil and flux residue, then the system is sanitized and flushed again so that no sanitizing solution is left for an occupant to drink. Water running clear proves the flush worked, not that the piping is disinfected. The disinfectant residual in the municipal supply is intended to protect water already in the mains, not to disinfect newly installed pipe. Inhibitor chemicals belong in closed heating loops and must never be introduced into potable lines.

Q49. Answer: C

Discovering unapproved work creates liability concerns. The plumber should document the finding, inform the owner, and recommend obtaining retroactive permits and inspections for the unapproved work to protect all parties.

Q50. Answer: B

Heating galvanized (zinc-coated) steel produces zinc oxide fumes which cause metal fume fever — flu-like symptoms. Requires LEV (local exhaust ventilation) and respiratory protection.

Scored under 70%?

The real exam draws from a much wider pool. The **306A Plumber Complete Question Bank** gives you every question on our platform with full explanations and key-concept summaries — built to close exactly the gaps this mock exposed.

Get the 306A Plumber Complete Question Bank →

Or keep practising free online at redsealquiz.ca/306a.html

© 2026 · redsealquiz.ca/306a.html · For personal study use only. Independent, unofficial study resource — not affiliated with or endorsed by the CCDA, the Red Seal Program, or any apprenticeship authority. See "About This Material".